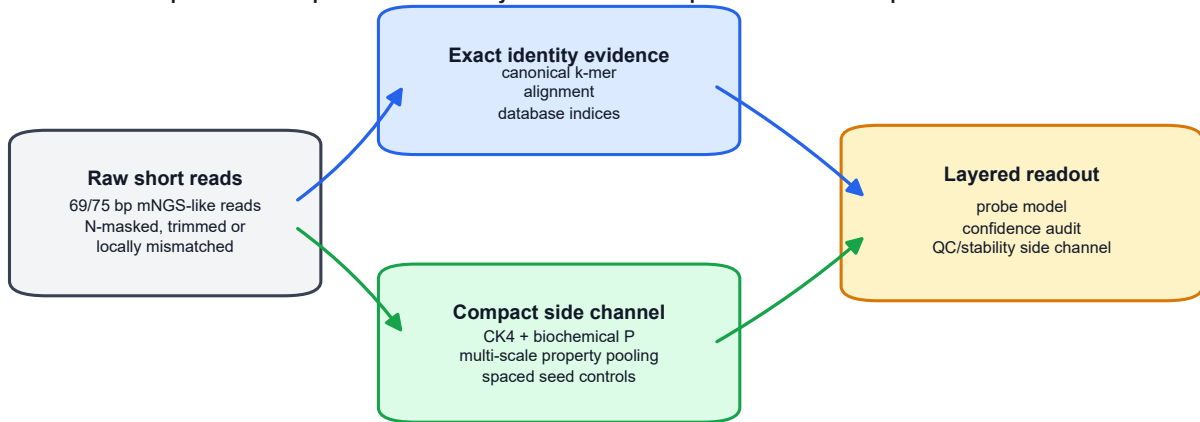


Short-read representation separates exact identity evidence from compact biochemical and position-aware side channels.



Full position encodings are used as upper-bound diagnostics, not as the primary clinical classifier.